

Confidence Intervals & Hypothesis Testing - Case Studies (Ch. 8-9)

All calculations shown with clear interpretations. Confidence level 95% unless stated.

Submission: Worked Solutions with Explanations

Below are fully worked solutions for three case studies covering confidence intervals and hypothesis testing. Each answer includes setup, calculations, decisions, and clear interpretations in context. Labeling follows the assignment questions exactly.

Case Study 1: Confidence Intervals in Business Decision Making

Given: A random sample of $n = 50$ stores has sample mean $\bar{x} = \$4,500$ and sample standard deviation $s = \$600$.

Q1. Construct a 95% confidence interval for the true mean monthly sales.

Because $n = 50$ is reasonably large, the normal (z) approximation is appropriate. Standard error (SE) = $s/\sqrt{n} = 600/\sqrt{50} = 84.85$. With $z^* = 1.96$ for 95% confidence, margin of error (ME) = $1.96 * 84.85 = 166.3$. $CI = \bar{x} \pm ME = 4500 \pm 166.3 = (4333.7, 4666.3)$.

Q2. Interpret the confidence interval in the business context.

We are 95% confident that the true average monthly sales of the new product line across all stores lies between \$4,334 and \$4,666. If the company were to repeat this sampling process many times, about 95% of similarly built intervals would contain the true mean.

Q3. Evaluate the marketing claim that the true mean is at least \$4,700.

The claim $\mu \geq 4700$ is not supported by our CI because 4700 is above the upper bound (4666.3). Based on this sample, the data do not justify claiming an average of at least \$4,700.

Q4. How to reduce the margin of error?

- Increase the sample size n (ME decreases with $1/\sqrt{n}$).
- Reduce variability via better measurement or more homogeneous store groups (lower s).
- Use a lower confidence level (trades precision for confidence, usually not preferred).

Case Study 2: Hypothesis Testing in Healthcare

Given: Prior mean time to pain relief $\mu_0 = 30$ minutes. New drug test on $n = 40$ patients yields $\bar{x} = 28$, $s = 5$. We test at $\alpha = 0.05$ whether the new drug provides faster relief.

Q1. State hypotheses.

$H_0: \mu = 30$ (no improvement). $H_a: \mu < 30$ (new drug is faster). Left-tailed one-sample t test.

Q2. Conduct the test (compute statistic and decision).

Test statistic: $t = (\bar{x} - \mu_0) / (s/\sqrt{n}) = (28 - 30) / (5/\sqrt{40}) = -2 / 0.7906 = -2.53$. Degrees of freedom

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~ 39. Critical value for one-sided $\alpha = 0.05$ is $t^* = -1.685$. Since $-2.53 < -1.685$, we reject H_0 at the 5% level. (p-value ~ 0.0076)

Q3. Interpretation in context.

There is statistically significant evidence that the new drug provides faster pain relief than the old drug. On average, time to relief is less than 30 minutes.

Q4. Possible error type and implications.

By rejecting H_0 , the relevant risk is a Type I error: concluding the new drug is faster when in truth it is not. Implication: adopting a drug without true improvement could misallocate resources and affect patient expectations. Conversely, if we had failed to reject H_0 when the drug truly is faster, that would be a Type II error.

Case Study 3: Proportion Testing in Public Opinion Research

Given: Of $n = 800$ citizens surveyed, $x = 480$ approve the policy, so $\hat{p} = 480/800 = 0.60$.

Q1. 95% confidence interval for the approval proportion.

$SE = \sqrt{\hat{p}(1 - \hat{p}) / n} = \sqrt{0.6 \cdot 0.4 / 800} = \sqrt{0.24/800} = 0.01732$. $ME = 1.96 \cdot 0.01732 = 0.0339$. $CI = 0.60 \pm 0.0339 = (0.566, 0.634)$.

Q2. Interpretation for decision-making.

We are 95% confident that 56.6% to 63.4% of citizens approve. This suggests majority support, providing evidence to proceed, while also planning for engagement with the 36% to 43% who may not approve.

Q3. Has approval decreased relative to 55%? ($\alpha = 0.05$)

$H_0: p = 0.55$ (no decrease). $H_a: p < 0.55$ (decrease). Left-tailed one-proportion z test.

Test statistic: $z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0) / n} = (0.60 - 0.55) / \sqrt{0.55 \cdot 0.45 / 800} = 0.05 / 0.01757 = 2.84$.

Because the test is left-tailed and $z = +2.84$ (which lies in the right tail), the p-value is large; we fail to reject H_0 . Data do not support a decrease; if anything, support appears higher.

Q4. Recommendations to government.

- Approval has not decreased; point estimate (60%) exceeds 55%.
- Communicate majority support while continuing outreach to dissenters.
- Monitor trends with follow-up surveys and stratified analyses (region, age) to detect shifts.
- Consider confidence intervals, not just point estimates, in policy decisions.

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Submission Checklist

- Each answer is labeled and includes setup, calculation, decision, and interpretation.
- Confidence intervals use 95% with $z = 1.96$ unless small-sample t is needed.
- Hypothesis tests specify H_0/H_a , α , test statistic, and decision rule/p-value.
- This PDF is ready to submit. Rename as FirstName_LastName_Week11_Assignment.pdf.